Project Design Phase-II
Data Flow Diagram & User Stories

Date	03 October 2026
Team ID	SWTID-2026-5297
Project Name	Script-Controlled ACL - Restrict Record Based on Field value
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flow within the Script-Controlled ACL system. It clearly shows how data enters and leaves the system, how user roles and record values are processed, and how access to records is controlled. It also represents the flow of data between users, the ACL mechanism, and the Institution Details table.

Example:(simplified)

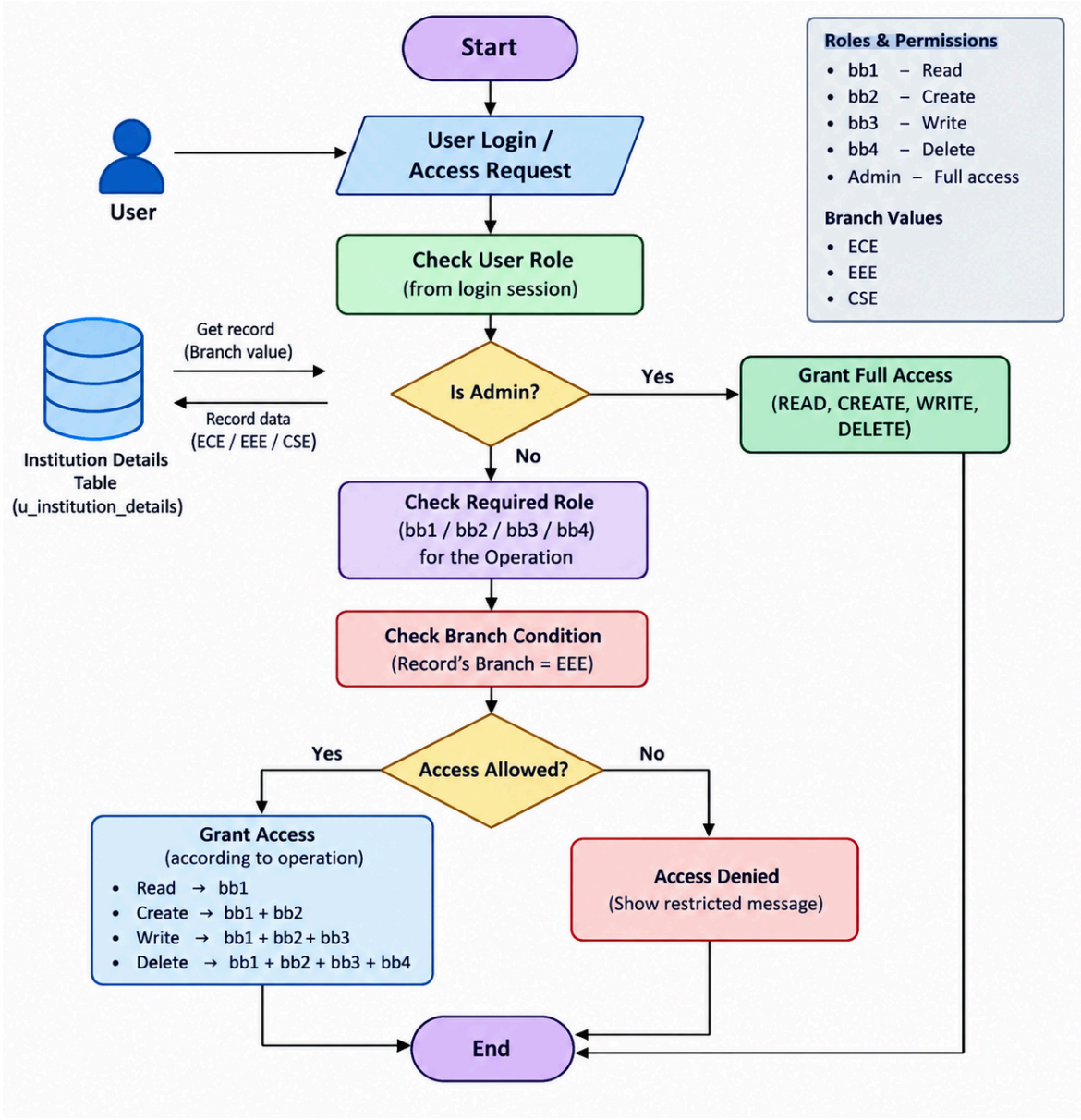

Start
User
User Login / Access Request
Check User Role (from login session)
Roles & Permissions
bb1 – Read
bb2 – Create
bb3 – Write
bb4 – Delete
Admin – Full access
Branch Values
ECE
EEE
CSE
Get record (Branch value)
Record data (ECE / EEE / CSE)
Institution Details Table (u_institution_details)
Is Admin?
Yes
Grant Full Access (READ, CREATE, WRITE, DELETE)
No
Check Required Role (bb1 / bb2 / bb3 / bb4) for the Operation
Check Branch Condition (Record's Branch = EEE)
Access Allowed?
Yes
No
Grant Access (according to operation)
Read → bb1
Create → bb1 + bb2
Write → bb1 + bb2 + bb3
Delete → bb1 + bb2 + bb3 + bb4
Access Denied (Show restricted message)
End

Diagram / Timeline – Description

The diagram shows the Script-Controlled ACL workflow in a simple sequence: User Login → Role Verification → Branch Condition Check → Access Decision → Access Granted / Access Denied. Admin users receive full access, while other users are given permissions based on their assigned roles and ACL conditions.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
EEE User	Record Access	USN-1	As an EEE user, I can access Institution Details records according to my assigned ACL role.	The system checks the user's assigned role before providing access.	High	Sprint-1
EEE User	Read Access	USN-2	As a user with the bb1 role, I can view the permitted EEE branch records.	The user can view the permitted EEE branch records.	High	Sprint-1
EEE User	Create Access	USN-3	As a user with bb1 and bb2 roles, I can create new Institution Details records.	The user can access the New option and create a record.	High	Sprint-1
EEE User	Write Access	USN-4	As a user with bb1, bb2 and bb3 roles, I can edit Institution Details records.	The user can view and edit the permitted EEE records.	High	Sprint-1
EEE User	Delete Access	USN-5	As a user with bb1, bb2, bb3 and bb4 roles, I can delete Institution Details records.	The user can view, create, edit and delete the permitted EEE records.	High	Sprint-2
Normal User	Access Restriction	USN-6	As a user without the required ACL role, I cannot access restricted records.	The system denies access when the required role is not available.	High	Sprint-1
Administrator	Full Access	USN-7	As an administrator, I can view all Institution Details records regardless of branch.	The administrator can view all records regardless of the Branch value.	High	Sprint-1
Administrator	Record Management	USN-8	As an administrator, I can manage Institution Details records.	Administrator access is not restricted by the normal branch-based ACL rule.	High	Sprint-2
ACL System	Branch Validation	USN-9	As the ACL system, I can check the Branch value while controlling record access.	The Branch value is evaluated during the access-control process.	High	Sprint-1
ACL System	Security Enforcement	USN-10	As the ACL system, I can allow or deny access according to the defined ACL rules.	Authorized users are allowed and unauthorized users are denied.	High	Sprint-2